Bolla Shiva Shankar

Phone: +91 9346213260

Warangal, Telangana

Email: shivabolla7@gmail.com

LinkedIn: https://www.linkedin.com/in/shiva-shankar-bolla-34724a257/

Github: https://github.com/ShivaBolla

CAREER OBJECTIVE

BTech graduate in Computer Science and Engineering with strong skills in Python, SQL and web development. Experienced in building real-time projects like a sign language recognition system and tourism website. Quick learner with good communication and teamwork abilities.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
BTech (CSE)	Bharath University	82%	2021-25
Intermediate (MPC)	Sri Chaitanya Junior College	84.2%	2019-21
S.S.C	St. Mary's High School	90.25%	2019

TECHNICAL SKILLS

- **Programming**: Python, Java.
- **Databases**: SQL, Mongodb.
- **Web Development**: HTML, CSS, JavaScript.
- **Tools/Technologies**: GitHub, VSCode, Oracle Database.

PROJECTS

1) **Sign Language Recognition System Using Machine Learning:** Developed a real-time Sign Language Recognition System using Machine Learning to accurately interpret American Sign Language (ASL) hand gestures and convert them into corresponding text.

Workflow

- **Data Collection:**
  Used Python script (data_collection.py) to collect gesture images with hand-tracking using MediaPipe & OpenCV. Each image was saved in label-based folders.
- **Labels Creation:**
  Stored labels in a text file (labels.txt) representing each ASL character/sign used for classification.
- **Model Creation:**
- Used Teachable Machine to train the model with the collected dataset.
- Exported trained model as keras_model.h5 to use in real-time recognition.
- **Testing/Prediction (test.py):**
- Used the trained model for live hand sign recognition.
- Displayed predicted sign using OpenCV's real-time video feed.

- Used CVZone and MediaPipe for hand detection and tracking.
- **Python Libraries Used:**
- OpenCV: For image processing and real-time video feed.
- CVZone: For easier hand detection and bounding box creation.
- MediaPipe: For hand landmark detection and gesture tracking.
- NumPy: For array and data manipulation.
- Math: To measure hand angles and distances.
- Time**:** Used in data collection timing and delays.
- **Outcome:**
- Successfully recognized and displayed ASL gestures in real-time.

Tools & Technologies:

- **Languages**: Python
- **Libraries**: OpenCV, CVZone, MediaPipe, NumPy, Math, Time
- **Other Tools**: Teachable Machine (Google)
- **Model File**: keras_model.h5 (Trained model file for ASL sign classification)
- **Dataset**: ASL (American Sign Language) Signs Dataset.

2) **Tourism Website:** Designed and developed a fully responsive tourism website showcasing Telangana's cultural heritage, historical sites, and major tourist destinations.
   - Built using HTML for structured content and CSS for visually appealing layouts.
   - Integrated JavaScript to add interactivity, including background theme toggling, dynamic marquee text color changes, and engaging animations for better user experience.
   - Implemented responsive design with media queries for smooth compatibility across mobile, tablet, and desktop devices.
   - **Technologies Used:** HTML, CSS, JavaScript, VS Code.

TRAINING

- Completed Python Full Stack training at QSpiders.
- Learned Python, SQL, HTML, CSS, JavaScript, Django with real-time projects and interview preparation.

ACHIEVEMENTS

1. Selected for QSpiders Incubation Program for Python Full Stack Development through campus placement.
2. Awarded "Best Student of the Year" and served as Sports & Group Leader.

CERTIFICATIONS

- Completed Python course from Udemy.
- Completed SQL from Great Learning.
- Completed Web Technologies from Great Learning.

STRENGTHS

- Positive Attitude.
- Good Communication Skills.